ICDFA ACADEMY

CIP-A107 · Security Operations Centre (SOC) Essentials
Laboratory Practical Assessment

FILE INTEGRITY MONITORING WITH WAZUH

SIEM Deployment, Agent Registration & Real-time FIM

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Course	CIP-A107 — SOC Essentials
Lab Focus	File Integrity Monitoring (Wazuh)
Analysis type	Hands-on SIEM Lab — Manager + Windows Agent
Environment	VirtualBox — Ubuntu Manager · Windows Agent
Classification	**TRAINING USE ONLY**

This report contains only evidence obtained inside the authorised laboratory environment.

1. Executive Summary

This report presents the results of a controlled laboratory exercise deploying Wazuh as a Security Information and Event Management (SIEM) platform and validating File Integrity Monitoring (FIM) on a Windows endpoint. All work was conducted inside a VirtualBox lab with bridged networking between an Ubuntu Manager and a Windows agent.

The Wazuh all-in-one stack (Manager, Indexer, Dashboard) was installed on Ubuntu. A Windows agent was enrolled using the manage_agents key-exchange workflow and confirmed Active on the dashboard. Real-time FIM was configured via the syscheck module against a dedicated test directory. Create, modify, and delete actions produced syscheck alerts (rules 550, 553, 554) that were visible under Integrity Monitoring and Security Events.

The laboratory objectives—manager deployment, agent registration, FIM configuration, alert generation, and SOC-style analysis—were achieved. Conclusions are supported by the screenshots and configuration evidence collected during the session.

1.1 Key Results at a Glance

Metric	Value	Significance
Platform	Wazuh 4.x	Open-source SIEM / XDR
Manager host	Ubuntu (VirtualBox)	All-in-one install
Agent host	Windows 10 / 11	Endpoint under monitoring
FIM mode	Realtime (syscheck)	Immediate create/modify/delete alerts
Monitored path	C:\FIM-Lab / SensitiveFiles	Lab test directory
Alert rules observed	550 · 553 · 554	Modified · Deleted · Added
Agent status	Active (green)	Confirmed on dashboard

2. Purpose and Scope

This laboratory exercise required deployment of a central Wazuh Manager, enrolment of a Windows endpoint agent, configuration of File Integrity Monitoring on a sensitive directory, generation of controlled file-change events, and analysis of the resulting alerts from a SOC analyst perspective.

All activities remained inside the authorised VirtualBox lab. The four analysis questions required by the assignment brief (alert triage, false-positive identification, ransomware use-case, and comparison with native Windows Event Logs) are answered in Section 11.

3. Lab Architecture and Topology

The environment consists of an Ubuntu virtual machine hosting the Wazuh stack and a Windows host (or Windows VM) running the agent. Bridged networking places both systems on the same Layer-2 segment.

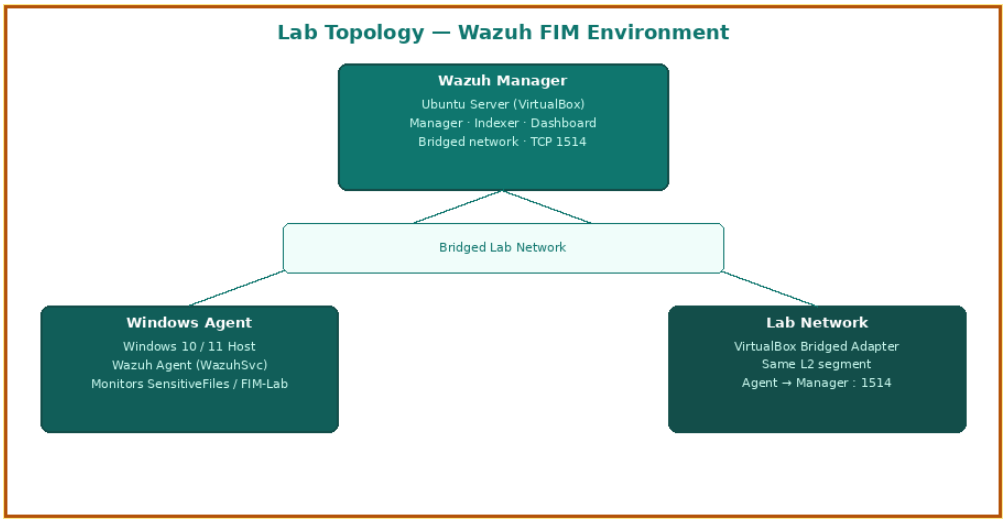


Figure 1 — Lab topology: Wazuh Manager (Ubuntu), Windows agent, bridged network.

Component	Host	Role
Wazuh Manager	Ubuntu (VirtualBox)	Collects, analyses, stores events
Wazuh Agent	Windows 10 / 11	Ships logs and FIM events
Network	Bridged Adapter	Agent ↔ Manager on TCP 1514

> **Note:** In production the Manager, Indexer and Dashboard are typically split across nodes. The all-in-one installer is appropriate for this learning lab.

4. Prerequisites

- Oracle VirtualBox installed on the physical machine.
- Ubuntu Server 20.04+ (or 22.04) VM with bridged networking and internet access.
- Administrative rights on the Windows endpoint that will run the agent.
- A modern web browser for the Wazuh dashboard.

5. Part 1 — Deploying the Wazuh Manager

All commands below were executed on the Ubuntu virtual machine from a terminal session.

5.1 Add the Wazuh GPG Key

The official GPG key is imported so the package manager can verify package authenticity:

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | \
  sudo gpg --dearmor -o /usr/share/keyrings/wazuh-archive-keyring.gpg
```

5.2 Download and Run the Installation Script

The official installer deploys Manager, Indexer and Dashboard in one pass. The -a flag installs every component; -i enables interactive prompts.

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && \
  sudo bash ./wazuh-install.sh -a -i
```

When the script finishes it prints the dashboard URL and the admin password. That password is required for the next step.

5.3 Service Management

```
sudo systemctl status  wazuh-manager
sudo systemctl restart wazuh-manager
sudo systemctl stop    wazuh-manager
sudo systemctl start   wazuh-manager
```

6. Part 2 — Accessing the Wazuh Dashboard

Discover the Ubuntu VM IP address:

```
ip addr show
```

Open a browser and navigate to https://<UBUNTU_VM_IP>. Accept the self-signed certificate warning (expected in a lab). Log in with username admin and the password printed by the installer.

> **Note:** This dashboard is the primary SOC console for agent health, alert triage and investigation.

7. Part 3 — Installing the Wazuh Agent (Windows)

On the Windows host the matching agent MSI was downloaded from the official packages repository (example for the 4.12 series):

```
https://packages.wazuh.com/4.x/windows/wazuh-agent-4.12.0-1.msi
```

The installer was run with default options. The WazuhSvc service is registered but remains inactive until a manager address and authentication key are supplied.

8. Part 4 — Agent–Manager Registration

Communication is authenticated with a unique key generated on the manager.

8.1 Generate the Agent Key

On the Ubuntu manager:

```
sudo /var/ossec/bin/manage_agents
```

Inside the interactive menu:

- Press A to add an agent.
- Assign a clear name (e.g., Windows-SOC-Lab).
- Leave the IP as any, or enter a static Windows IP.
- Confirm, then press E to extract the key.
- Copy the long base64 key for the next step.

8.2 Apply the Key on Windows

Wazuh Agent Manager was opened from the Start menu. On the Management tab the key was pasted, the Ubuntu VM IP entered as the manager address, Apply clicked, then Restart.

Within approximately one minute the agent appeared under Agents on the dashboard with a green Active status.

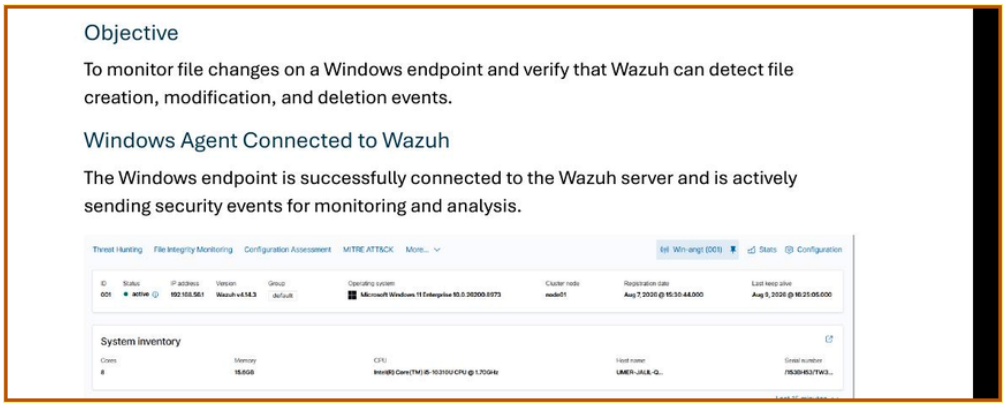


Figure 2 — Windows agent shown as Active in the Wazuh dashboard with system inventory and event summary.

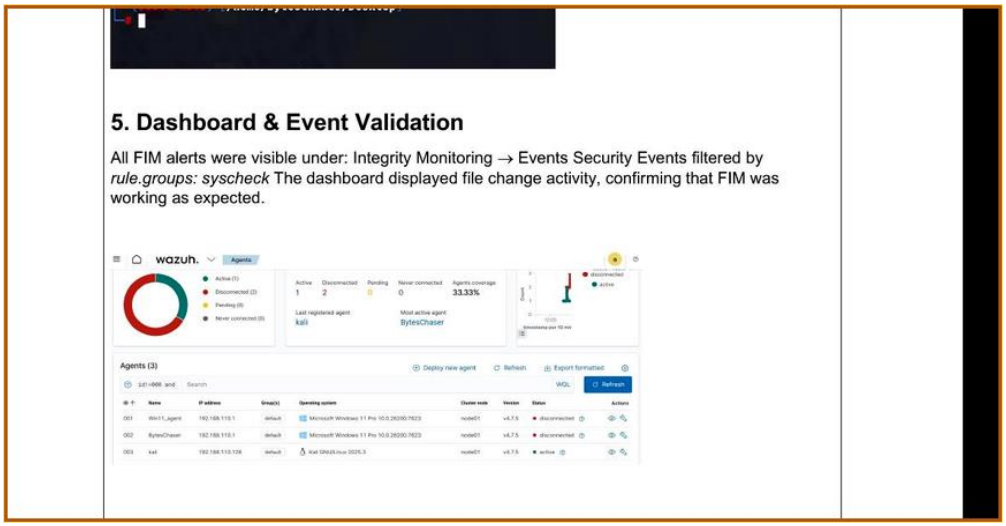


Figure 3 — Agents list confirming enrolled endpoints with Active status.

9. Part 5 — Implementing File Integrity Monitoring

FIM is provided by the syscheck module. Real-time mode watches the directory continuously and raises an alert as soon as a file is created, modified or deleted.

9.1 Create the Monitored Directory

A dedicated test folder was created on the Windows endpoint (C:\FIM-Lab or C:\Users\<Username>\SensitiveFiles).

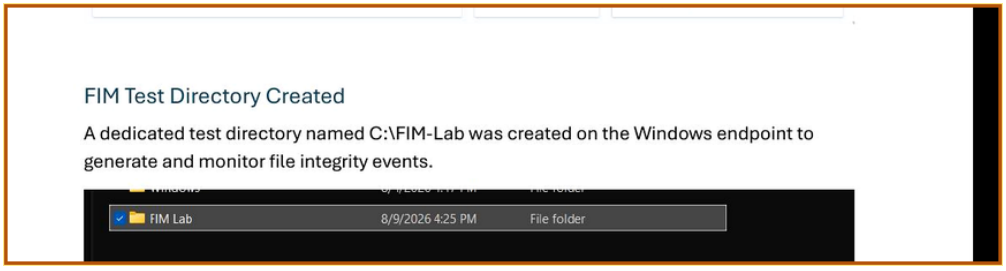


Figure 4 — Test directory created on the Windows endpoint for FIM validation.

9.2 Edit ossec.conf

The agent configuration was opened as Administrator:

```
C:\Program Files (x86)\ossec-agent\ossec.conf
```

Inside the <syscheck> block the following directory entry was added:

```
<directories realtime="yes">C:\FIM-Lab</directories>


<!-- richer attributes (optional) -->
<directories check_all="yes" report_changes="yes" realtime="yes">
  C:\Users\YourUsername\SensitiveFiles
</directories>
```

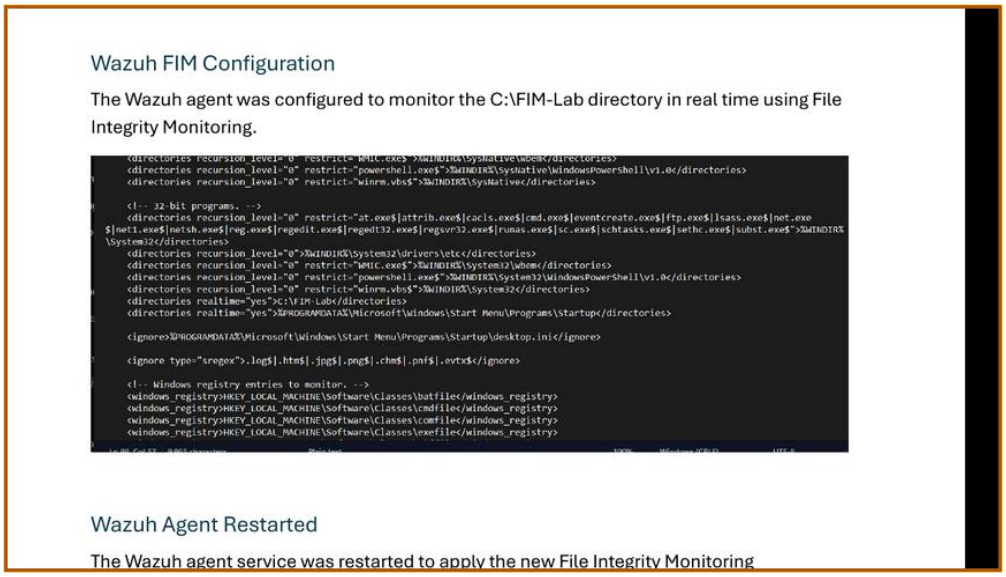


Figure 5 — Wazuh agent ossec.conf showing real-time FIM directory configuration on Windows.

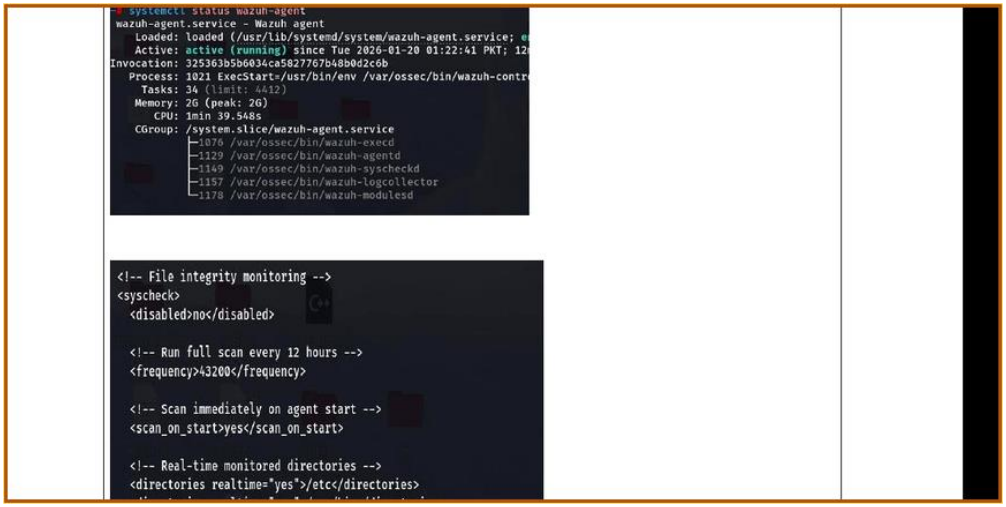


Figure 6 — syscheck section with realtime="yes" enabled for monitored paths.

9.3 Restart the Agent

The configuration was applied by restarting the service from an elevated PowerShell prompt:

```
Restart-Service WazuhSvc
Get-Service WazuhSvc
```

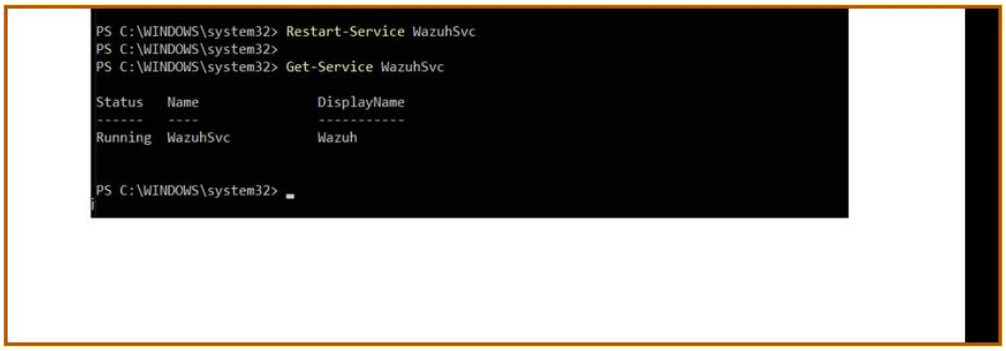


Figure 7 — Wazuh agent service restarted successfully after the configuration change.

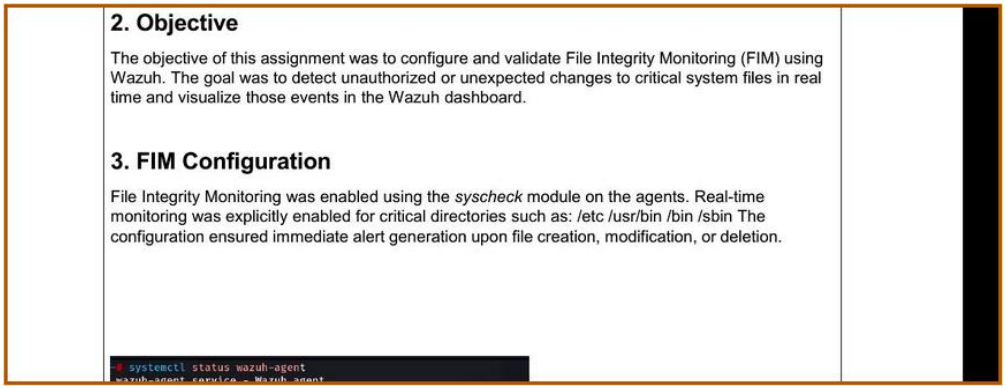


Figure 8 — Agent service status confirming the process is active and running.

10. Part 6 — SOC Verification and Alert Analysis

Test events were generated inside the monitored folder to validate detection.

10.1 Generate Test Events

1. Create a new text file (e.g., secret_plan.txt or fim_test.txt).
2. Modify the file by adding content and saving.
3. Delete the file.



Figure 9 — File creation, modification and deletion commands used to trigger FIM alerts.

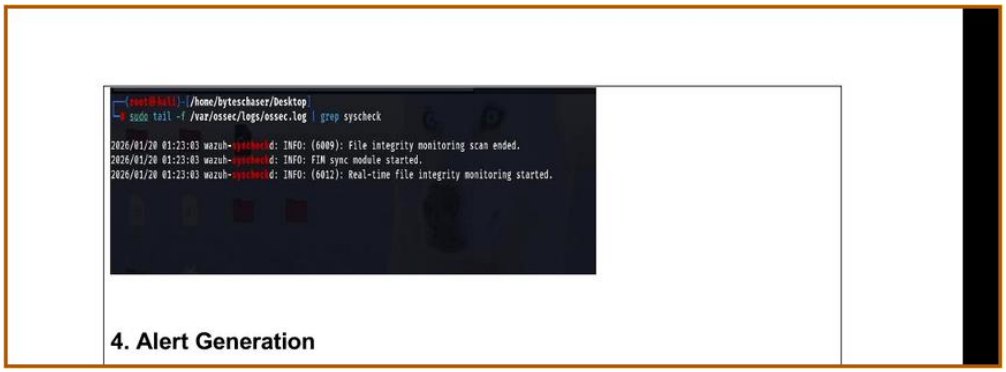


Figure 10 — Local agent log confirming FIM scan and real-time monitoring start.

10.2 Investigate Alerts

In the Wazuh dashboard, Security Events and Integrity Monitoring were opened. The filter rule.groups: "syscheck" isolated FIM activity. Each alert includes the full path, action (added / modified / deleted), timestamp, agent name and cryptographic hashes.

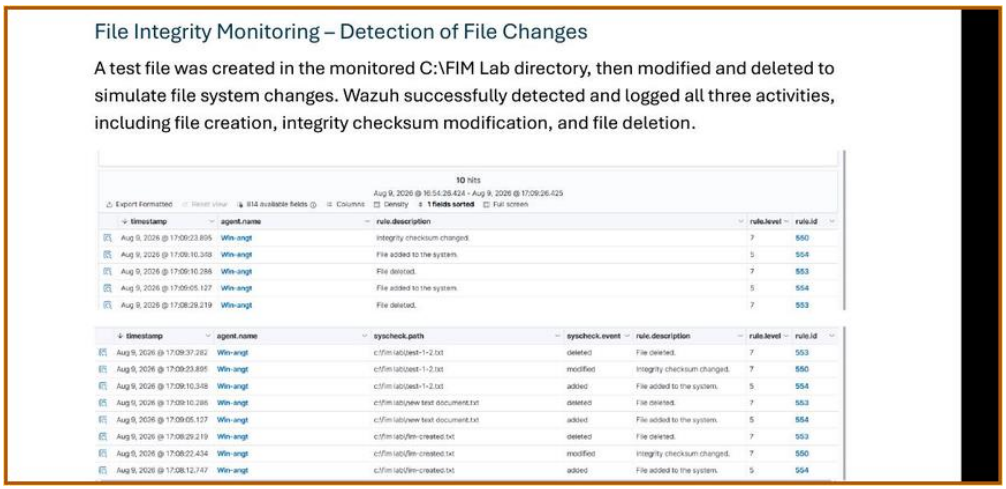


Figure 11 — FIM alerts: file added, integrity checksum changed and file deleted (rules 550 / 553 / 554).

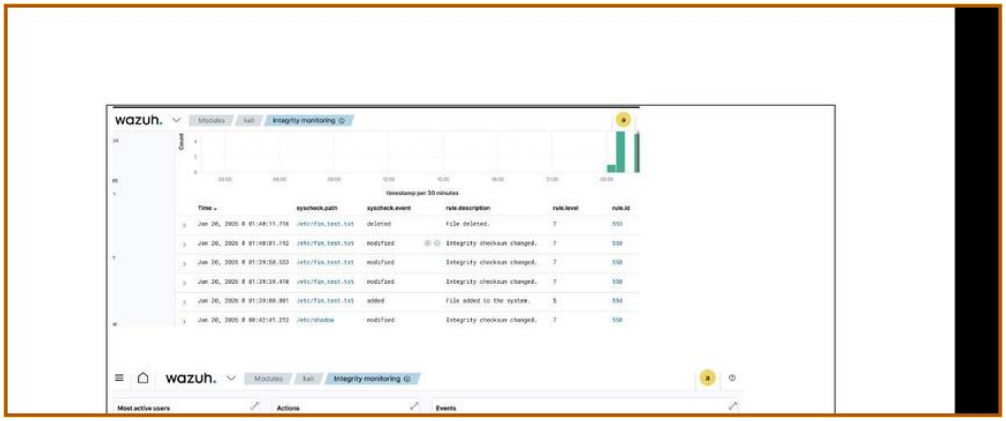


Figure 12 — Integrity Monitoring events table showing syscheck paths and rule descriptions.



Figure 13 — Integrity Monitoring dashboard: actions breakdown, event timeline and file-change summary.

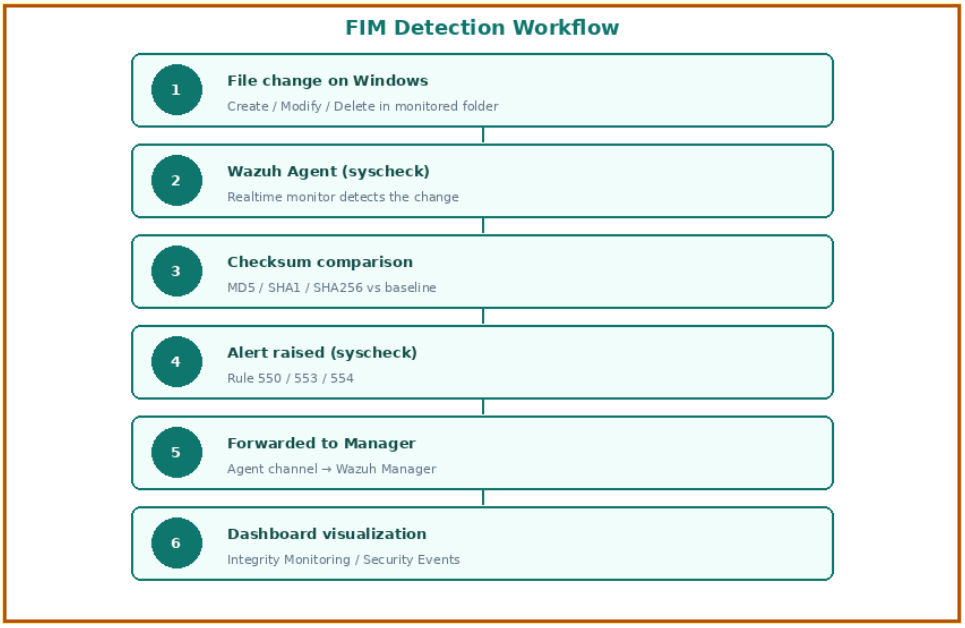


Figure 14 — End-to-end FIM detection workflow from file change to dashboard alert.

11. Lab Challenge and Analysis Questions

Q1 — Alert Triage

Question: Find an alert where a file was modified. What specific details tell you what changed about the file?

Answer: A modification alert (typically rule ID 550 — “Integrity checksum changed”) carries the forensic attributes required for triage:

- syscheck.md5_before / md5_after — previous and current MD5 hashes.
- syscheck.sha1_before / sha1_after and sha256 equivalents — stronger fingerprints.
- syscheck.size_before / size_after — size change in bytes.
- syscheck.mtime — modification timestamp.
- When report_changes is enabled, a text diff of the actual content change can also appear.

These fields prove that content or metadata actually changed, not merely that a process touched the file handle.

Q2 — False Positive Identification

Question: In a corporate environment, which folders would generate too many FIM alerts to be useful? Name two and explain why.

Answer:

- Temporary and cache directories (%TEMP%, C:\Windows\Temp, browser caches). Applications constantly create and delete short-lived files here, producing a continuous stream of low-value alerts.
- High-churn user paths such as Desktop or Downloads when users save drafts every few minutes. Without exclusions the SOC queue fills with legitimate activity.

Best practice is to limit real-time FIM to high-value locations (system binaries, configuration directories, application install paths, and a small set of sensitive data folders) and to define ignore rules for known noisy paths.

Q3 — Proactive Defense Against Ransomware

Question: How could you use Wazuh's FIM capability to monitor for a specific ransomware behavior?

Answer: Ransomware typically encrypts large numbers of user files in a short window and often drops a ransom note (README.txt, HOW_TO_DECRYPT.txt, etc.). A practical FIM approach includes:

- Real-time monitoring of high-value document locations (Documents, Desktop, shared drives).
- Detection of mass modification events — many files changing hashes within minutes — which can be correlated with custom rules.
- Alerting on the sudden appearance of known ransom-note filenames inside monitored directories.
- Optional integration with VirusTotal or YARA so newly created executables are automatically checked for malware signatures.

Early detection of the first wave of encrypted files or the ransom note gives the SOC a chance to isolate the host before encryption completes.

Q4 — Tool Comparison

Question: How does a centralized tool like Wazuh for FIM provide an advantage over relying solely on built-in Windows features such as Event Logs?

Answer: Native Windows Object Access auditing and the Security event log can record file activity, but they have practical limits:

- Scale — reviewing Event Viewer on each endpoint does not scale beyond a handful of machines.
- Context — Windows events rarely include cryptographic hashes or a clear before/after picture.
- Centralization — Wazuh aggregates every agent's FIM events into one searchable index with dashboards, filters and alerting.
- Cross-platform coverage — the same policy and console cover Windows, Linux and macOS.

- Enrichment — MITRE ATT&CK mappings, compliance labels (PCI-DSS 11.5, NIST SI-7) and threat-intelligence integrations.

Windows Event Logs remain useful for local troubleshooting; Wazuh turns the same class of data into a SOC-ready detection and investigation capability.

12. Conclusion

The laboratory successfully demonstrated a complete FIM use-case on a Windows endpoint. The Wazuh Manager was deployed on Ubuntu, a Windows agent was enrolled and shown Active, real-time monitoring of a sensitive folder was enabled, and create / modify / delete actions produced clear syscheck alerts (rules 550, 553, 554) visible in the Integrity Monitoring and Security Events views.

The four analysis questions reinforce the practical value of FIM for alert triage, noise reduction, ransomware detection and centralized visibility. This foundation can be extended with who-data auditing, report_changes, centralized agent configuration and VirusTotal integration for automated malware checks on newly created files.

— End of Report —